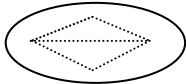


1(a)	histones ; A basic / histone, proteins	1
1(b)	<i>max 2 if the spindle / centrioles extend outside the cell</i> 1 two chromosomes, each with two chromatids, drawn approximately along central equator area ; R if drawn as a, bivalent / homologous pair (close together or further apart) A if equator is drawn across wider diameter 2 plus one other drawn feature ; e.g. spindle  <i>diamond/oval shape, minimum three spindle fibres as shown</i> or spindle fibre from each pole connecting to centromere area of one chromosome or centromere drawn holding sister chromatids together or <u>pair</u> of centrioles at each pole 3 any three labels ; I label to chromosome e.g. (sister / identical) chromatid/s centromere spindle / spindle fibres spindle equator / metaphase plate A equator <i>if spindle / spindle fibre label given</i> centriole/s A centrosome(s) / pole(s) <i>annotations may contain more than one acceptable label</i> <i>label lines to touch structure</i>	3

Question	Answer	Marks
1(c)	<i>assume sequential interphase to prophase if not named</i> any three from: <i>interphase max 2</i> (chromosomal) DNA replicates ; A DNA synthesis I chromatin R if stated as in, G1 / G2 (each new molecule of) DNA, complexes / associates / AW, with, histones / proteins ; sister / identical, chromatids form, qualified ; e.g. after (DNA) replication /during / end of, S phase <i>must be clear they are formed before prophase</i> <i>prophase max 2</i> chromatids / chromosome, condense ; R if also in prophase A become(s) shorter and fatter A coiling / supercoiling, <i>in context of</i>, DNA / chromosomes / chromatids R chromosomes condense to form chromatin becomes visible / appears, as two (sister / identical) chromatids ; I composed of two chromatids becomes attached to the spindle fibres ; (spindle) microtubules attach to centromere / kinetochore formation ;	3